

EXPLORATORY DATA ANALYSIS (EDA) ON TITANIC DATASET

Data Science & Analytics Internship

CodeBase Studio

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1. Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Titanic dataset to understand its structure, identify trends and patterns, detect missing values, validate assumptions through visualization, and gain insights for further analysis.

2. Dataset Description

The Titanic dataset contains information about passengers who traveled on the Titanic. The dataset includes demographic information, passenger class, fare, ticket details, age, gender, and survival status.

Total Records: 891

Total Variables: 12

Variables:

- PassengerId
- Survived
- Pclass
- Name
- Sex
- Age
- SibSp
- Parch
- Ticket
- Fare
- Cabin
- Embarked

3. Meaningful Questions Before Analysis

1. What factors influenced passenger survival?
2. Did gender affect survival chances?
3. What was the age distribution of passengers?
4. Are there missing values in the dataset?
5. Which passenger groups were more likely to survive?

4. Dataset Preview

```
df = pd.read_csv('Titanic-Dataset.csv')
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Explanation

The `df.head()` function displays the first five records of the Titanic dataset. It provides an initial understanding of the variables available, including passenger details, age, gender, ticket information, fare, and survival status.

5. Data Structure Exploration



`df.info()`

```
... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId     891 non-null   int64
1   Survived        891 non-null   int64
2   Pclass          891 non-null   int64
3   Name            891 non-null   object
4   Sex             891 non-null   object
5   Age            714 non-null   float64
6   SibSp           891 non-null   int64
7   Parch          891 non-null   int64
8   Ticket          891 non-null   object
9   Fare            891 non-null   float64
10  Cabin           204 non-null   object
11  Embarked        889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

Explanation

The dataset contains 891 records and 12 variables. It includes both numerical and categorical variables. Numerical variables include PassengerId, Survived, Pclass, Age, SibSp, Parch, and Fare, while Name, Sex, Ticket, Cabin, and Embarked are categorical variables. The output also highlights missing values in some columns.

6. Missing Value Analysis

```
df.isnull().sum()
```

```
...
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age          177
SibSp         0
Parch         0
Ticket         0
Fare          0
Cabin       687
Embarked       2
dtype: int64
```

Explanation

The missing value analysis shows:

- Age: 177 missing values
- Cabin: 687 missing values
- Embarked: 2 missing values

The remaining columns contain complete information. These missing values should be handled during data preprocessing to improve data quality and analysis accuracy.

7. Statistical Summary

df.describe()

...	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

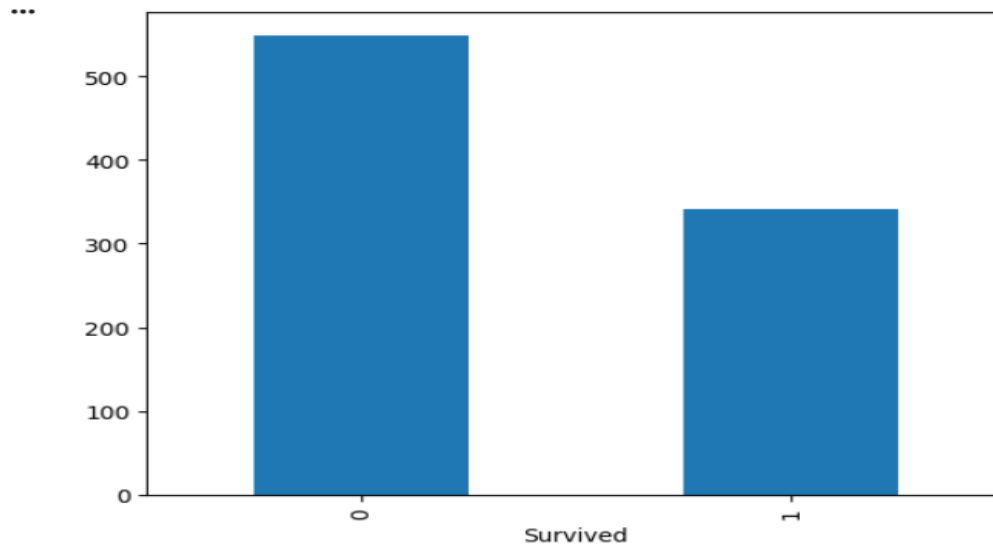
Explanation

Descriptive statistics provide a summary of the dataset. The average passenger age is approximately 29.7 years, while the average fare paid is 32.20. The maximum passenger age is 80 years, and the minimum age is 0.42 years. These statistics help understand the distribution and characteristics of the data.

8. Trend and Pattern Analysis

8.1 Survival Analysis

```
df['Survived'].value_counts().plot(kind='bar')  
plt.show()
```



Explanation

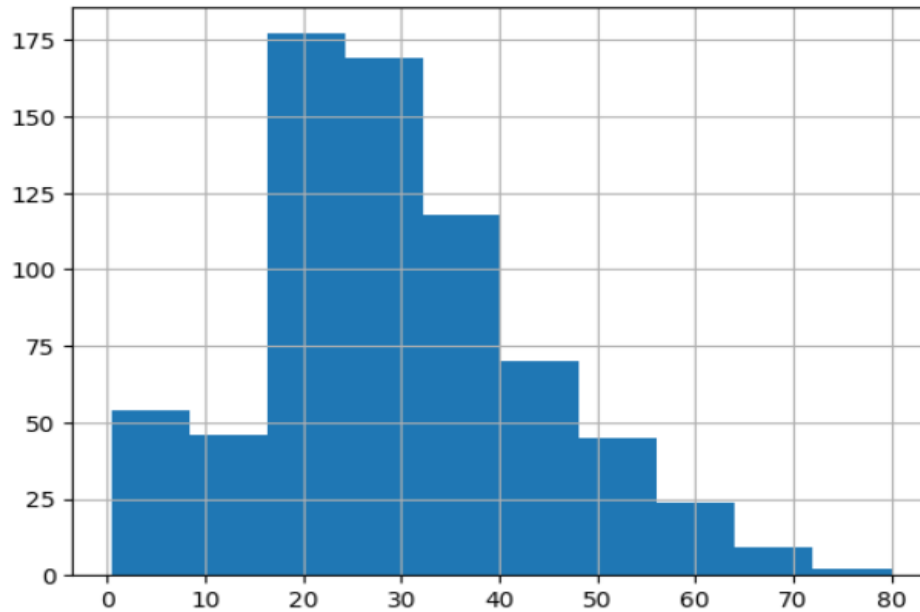
The survival graph shows that the number of passengers who did not survive was greater than the number of passengers who survived. This indicates that the majority of passengers lost their lives during the Titanic disaster.

8.2 Age Distribution Analysis



```
df['Age'].hist()  
plt.show()
```

...

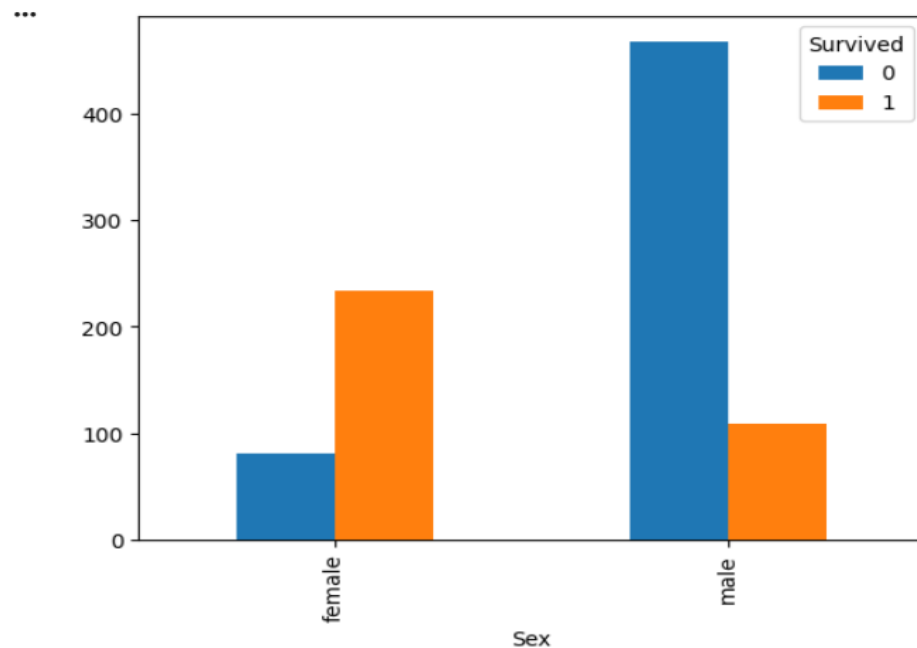


Explanation

The age distribution graph shows that most passengers were between 20 and 40 years old. The number of passengers gradually decreases for higher age groups, indicating that elderly passengers were fewer in number.

8.3 Gender and Survival Analysis

```
pd.crosstab(df['Sex'],df['Survived']).plot(kind='bar')  
plt.show()
```



Explanation

The graph demonstrates a strong relationship between gender and survival. Female passengers had a significantly higher survival rate compared to male passengers. Male passengers experienced a higher number of deaths during the disaster.

9. Trends, Patterns and Anomalies

Based on the analysis, the following patterns were identified:

- More passengers died than survived.
- Female passengers had a higher survival rate than male passengers.
- Most passengers were between 20 and 40 years old.
- A large number of values are missing from the Cabin column.
- The Age column also contains several missing values.

These observations reveal important patterns within the dataset and highlight areas requiring further preprocessing.

10. Hypothesis Testing and Validation

Hypothesis 1

Female passengers were more likely to survive than male passengers.

Result

Accepted.

Evidence

The Gender vs Survival visualization clearly shows that female passengers had a much higher survival count compared to male passengers.

Hypothesis 2

Most passengers were adults.

Result

Accepted.

Evidence

The Age Distribution histogram indicates that the majority of passengers were between 20 and 40 years old.

11. Data Issues and Problems

The following issues were identified during the analysis:

1. Age contains 177 missing values.
2. Cabin contains 687 missing values.
3. Embarked contains 2 missing values.
4. Missing values may affect model accuracy.
5. Data cleaning and preprocessing are required before machine learning analysis.

12. Conclusion

Exploratory Data Analysis was successfully performed on the Titanic dataset. The analysis revealed important insights regarding passenger demographics and survival patterns. Female passengers showed higher survival rates than males, and most passengers were adults between 20 and 40 years old. The dataset also contained missing values in the Age, Cabin, and Embarked columns, which should be addressed before advanced analysis. Overall, the EDA process provided valuable understanding of the dataset and prepared it for further statistical and predictive analysis.